

RUSHIKESH SHASHIKANT KADEPURKAR

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EDUCATION

COEP Technological University, Pune

Expected 2027

B.Tech in Computer Engineering | CGPA: 7.98 / 10

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, SQL

Frameworks & Libs: React.js, Node.js, FastAPI, Spring Boot, LangChain, LangGraph

Developer Tools: Docker, Nginx, Git, GitHub Actions, Linux, Redis, Celery

Databases: PostgreSQL, MySQL, MongoDB, Qdrant (Vector DB)

Core Concepts: DSA, System Design, OOP, OS, Networks, DBMS, Machine Learning, RAG, Agentic AI

INTERNSHIP

AI Intern | Indian Institute of Technology (IIT) Jammu

Jun 2025 – Jul 2025

- Architected an autonomous multi-agent system using **LangGraph** to conduct evidence-backed bug root-cause analysis across enterprise codebases.
- Constructed an AST code chunking pipeline using **Tree-sitter** and **Qdrant** for context-aware semantic retrieval; decoupled repository indexing via **Celery** and **Redis** background workers.
- Engineered production-ready **FastAPI** services delivering token-streaming AI responses with sub-second latency.

TECHNICAL PROJECTS

NAT Traversal: UDP Hole Punching over QUIC

Feb 2026 – Apr 2026

C++, Python, aioquic, pybind11, WebSockets, STUN, TLS 1.3

- Engineered a P2P NAT traversal engine handling all 4 NAT configurations using STUN-based dual UDP probing.
- Built a high-frequency packet burst engine via native **C++ (pybind11)**, bypassing Python GIL to achieve 200ms retry cycles with seamless WebSocket fallback for symmetric NATs.
- Integrated **QUIC** protocol via aioquic featuring 0-RTT connection resumption, slashing reconnection overhead by 50–100ms with multiplexed streams for chat, files, and metrics.

Event-Sourced Bank Ledger

Aug 2025 – Sep 2025

Java, Spring Boot, PostgreSQL, CQRS, Event Sourcing, REST API

- Implemented high-concurrency ledger processing using **CQRS** and **Event Sourcing** paradigms in Spring Boot.
- Designed an immutable append-only event store in **PostgreSQL** with optimistic locking, eliminating balance race conditions and supporting atomic multi-aggregate transfers.
- Engineered an asynchronous projection pipeline rebuilding normalized read-models on demand from raw event logs to ensure strict eventual consistency.

SwiftTalk: Real-Time AI Messaging Platform

Sep 2025 – Oct 2025

React.js, Node.js, MySQL, WebSocket, LangChain, Docker, Nginx, CI/CD

- Developed an end-to-end collaborative messaging platform supporting group rooms, presence indicators, read receipts, and secure JWT/OTP session management.
- Integrated **LangChain** to drive automated chat translation, smart predictive replies, and conversational thread summarization.
- Containerized services using **Docker** and established automated testing/deployment pipelines via **GitHub Actions** behind an **Nginx** reverse proxy.

CERTIFICATIONS

Deep Learning – NVIDIA • **Backend Development** – IBM • **Intermediate to Advanced Python** – Udemy

EXTRACURRICULAR

Volunteer | Shantivan NGO

May 2024 – Jul 2024

Mentored orphan children in basic literacy and computer skills, fostering foundational knowledge and digital confidence. Designed interactive educational activities to enhance engagement and retention. Promoted environmental awareness programs through hands-on workshops and community clean-up drives.